$$\text{New Amount} = \text{Original Amount} + \frac{r\%}{100}\text{Original Amount}$$

In 2010:

Total swimmers = 35

a = number of advanced swimmers

b = number of intermediate swimmers

From 2010 to 2020:

Advanced swimmers increased by 53%

Intermediate swimmers increased by 44%

Total swimmers increased by 49%

Equation?

Advanced swimmers

Original number (2010)$= a$, Increase percent $= 53\%$

New number (2020)$= a + \frac{53}{100}a = 1a + 0.53a = 1.53a$

Intermediate swimmers

Original number (2010)$= b$, Increase percent $= 44\%$

New number (2020)$= a + \frac{44}{100}a = 1a + 0.44a = 1.44a$

Total swimmers in 2020

Original number (2010) $= 35$, Increase percent $= 49\%$

New number (2020) $= 35 + \frac{49}{100}(35) = 1(35) + 0.49(35) = 1.49(35)$

in 2020

advanced swimmers	+	intermediate swimmers	= Total swimmers
$1.53a$	+	$1.44b$	$= 35(1.49)$